



FRANKONIA

EMC Test-Systems GmbH

EMCL-35

EM-Coupling clamp
DATA SHEET



Features:

- ✓ For the coupling during immunity tests according to IEC/EN 61000-4-6
- ✓ Frequency range 10 kHz – 1 GHz
- ✓ Suitable for cables up to 35 mm, for thinner cables up to 20 mm: use CDN EMCL-20
- ✓ For cable diameter up to 35 mm, calibration unit and calibration data are supplied with each unit!

Frankonia EMC Test-Systems GmbH

Daimlerstr. 17

91301 Forchheim

Germany

+49 9191 73666-0

+49 9191 73666-20

sales@frankonia-emv.com

WWW.FRANKONIA-SOLUTIONS.COM



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OVERVIEW

EM coupling clamps are always used when the use of coupling / decoupling networks (CDN) does not seem suitable or appropriate. This applies, for example, to unshielded cables consisting of a large number of individual conductors.

With the electromagnetic coupling path, both capacitive and inductive coupling into the cable connected to the test object is achieved. The EM coupling clamp has a directivity of ≥ 10 dB above 10 MHz, so that a defined impedance between the asymmetrical connection point of the auxiliary / additional equipment and the reference ground plane is no longer required.

Above 10 MHz, the behavior of the EM coupling link is therefore similar to that of coupling/decoupling networks.

KEY FACTS

- ✓ EM-coupling clamps for immunity tests on cables up to 35 mm diameter
- ✓ Low insertion loss: less than 15 W amplifier power is required to achieve the highest test level of 10 V
- ✓ Supplied as standard with calibration set and individual calibration data

TECHNICAL DATA

Frequency range	10 kHz – 1 GHz
Nominal impedance	50 Ohm
Max. input level	
0.01 (10 kHz) – 100 MHz	100 W, 15 min
100 – 230 MHz	100 W, 5 min
230 – 1000 MHz	50 W, 3 min
Connector	N-type female
Cable diameter max.	< 35 mm
Dimensions (L x W x H)	665 x 160 x 135 mm
Weight	approx. 14 kg



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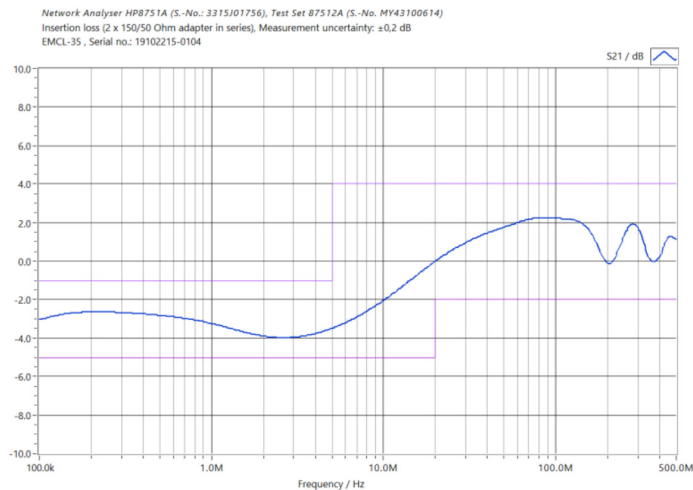
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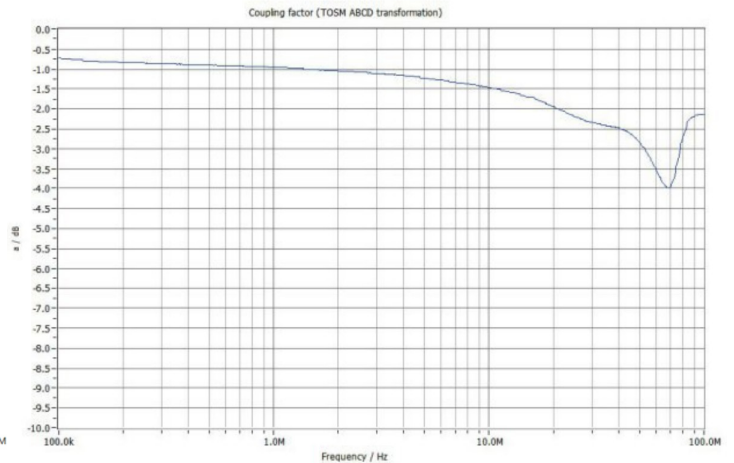


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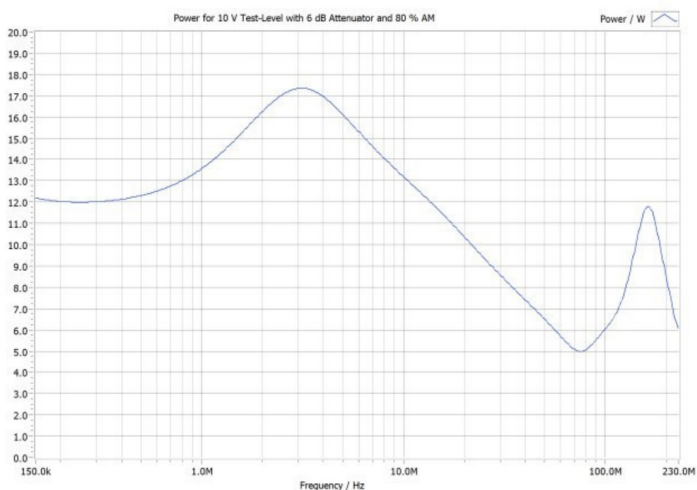
TYPICAL COUPLING FACTOR



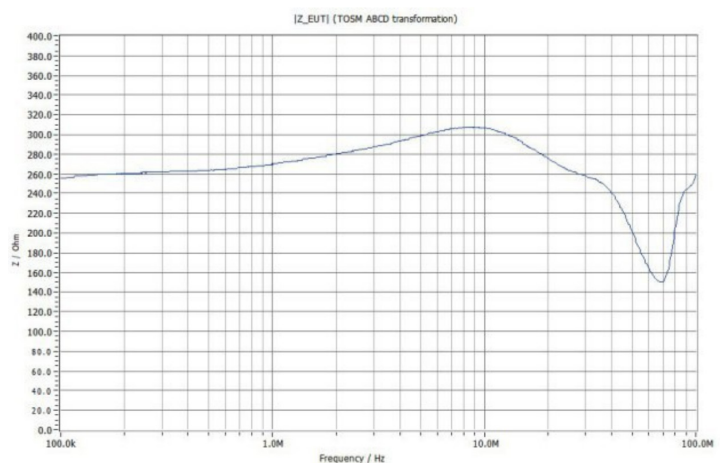
TYPICAL DECOUPLING FACTOR (TRANSFORMED)



MEASURED AMPLIFIER POWER



TYPICAL CLAMP IMPEDANCE (TRANSFORMED)



Measured amplifier power to achieve the highest level required in EN 61000-4-6 of 10 V.

Measurement was carried out with 6 dB attenuator and 80% amplitude modulation.



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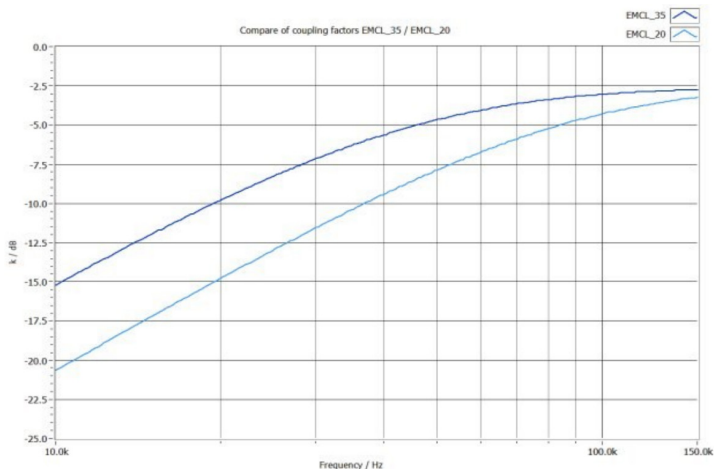


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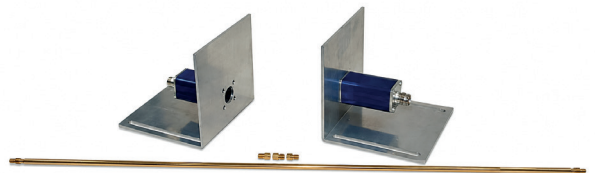
COMPARISON OF COUPLING FACTORS EMCL 35 / 20



The high coupling factor of the EMCL-35 allows a test according to level 3 (10V) of IEC 61000-4-6 already at 10 kHz with the test generator CDG 7000-75-10.

CALIBRATION SET / CALIBRATION DATA (INCLUDED IN DELIVERY)

- 2 mounting brackets incl. 50/150 Ohm transition
- 1 piece BNC termination, 50 Ohm, 1 W
- 1 adapter with banana plugs for calibration of the network analyzer, length 34 mm
- 1 piece brass rod 4 mm with banana plugs for setting the test level, length 672 mm
- N male to BNC female



CALIBRATION SET / CALIBRATION DATA (INCLUDED IN DELIVERY)

CDG 7006-20W	Attenuator 6 dB, 20 W, BNC connector male / female
CDG 7006-100W	Attenuator 6 dB, 100 W, BNC connector male / female
CDN ABCL-20	Decoupling clamp for additional decoupling during immunity testing according to IEC / EN 61000-4-6, 100 kHz - 1000 MHz, cable diameter max. 20 mm.
CDN ABCL-35	Decoupling clamp for additional decoupling during immunity testing according to IEC / EN 61000-4-6, 100 kHz - 1000 MHz, cable diameter max. 35 mm.